IDH3B-Related Retinopathy (Retinitis Pigmentosa 46, RP46): Comprehensive Disease Report

Disease: IDH3B-Related Retinopathy · **Causal gene:** *IDH3B* · **MONDO:** MONDO:0800393 · **OMIM:** #612572 (phenotype); *604526 (gene)* · *Category:** Mendelian (autosomal recessive)

Summary

IDH3B-related retinopathy is a rare, autosomal-recessive, nonsyndromic rod–cone dystrophy (retinitis pigmentosa) caused by biallelic loss-of-function variants in *IDH3B*, the gene encoding the β (structural/regulatory) subunit of mitochondrial NAD^+-specific isocitrate dehydrogenase (NAD-IDH, also called IDH3). The disease is catalogued as **Retinitis Pigmentosa 46 (RP46)**. Although the enzyme IDH3 is a housekeeping component of the tricarboxylic acid (TCA / Krebs) cycle present in every mitochondrion, its loss produces disease confined to the retina. This paradox is the central mechanistic story of the condition: in most tissues the NADP-dependent isocitrate dehydrogenase (IDH2) can substitute for the oxidative decarboxylation of isocitrate, but the metabolically extreme, mitochondria-rich photoreceptors of the retina have a particular requirement for NAD-IDH, so they alone degenerate when it fails.

The disease was discovered by **Hartong et al. (2008, *Nature Genetics*)** in two consanguineous families, in whom affected individuals were homozygous for loss-of-function *IDH3B* mutations (PMID: 18806796). Patient-derived cells showed NAD-IDH activity reduced to roughly 5% of normal, with an approximately **300-fold increase in the Km for NAD**, while NADP-IDH (IDH2) activity was preserved and patients had no extra-ocular disease. *IDH3B* is one of three subunit genes of the IDH3 heterotetramer, and strikingly **all three — *IDH3A*, *IDH3B*, and *IDH3G* — cause inherited retinal degeneration**, defining a convergent "IDH3-opathy." Within this family, *IDH3B* disease tends to be **milder and later-onset (≈30–50 years)** than *IDH3A* disease (childhood onset), a difference traced to the biochemistry of subunit assembly: αγ-dimers that survive loss of the β-subunit retain more catalytic activity than αβ-dimers.

This condition is extremely rare. Across ~546 RP probands screened in the original study, only two families carried *IDH3B* variants (~0.4%), and only a handful of pathogenic alleles have ever been reported worldwide: a frameshift **c.589delA (p.Ile197fs)**, a missense **p.Leu98Pro**, and a later missense **p.Gly286Glu**. There is no disease-specific treatment; management is supportive and follows general RP care (low-vision support, vitamin A considerations, surveillance for cataract and cystoid macular edema), with genetic counseling for families. No mouse model yet reproduces the retinopathy — *Idh3b*-null mice instead develop male infertility — which is itself informative about the residual-activity buffering that makes β-subunit loss comparatively mild.

Key Findings

Finding 1 — Biallelic *IDH3B* loss-of-function mutations cause autosomal-recessive RP (RP46)

The foundational observation, from **Hartong et al. 2008** (PMID: 18806796), identified two families with retinitis pigmentosa in which affected members were homozygous for loss-of-function mutations in *IDH3B*. The abstract states directly: *"Affected family members were homozygous for loss-of-function mutations in IDH3B, encoding the beta-subunit of NAD-specific isocitrate dehydrogenase (NAD-IDH, or IDH3), which is believed to catalyze the oxidation of isocitrate to alpha-ketoglutarate in the citric acid cycle."* The inheritance is autosomal recessive, and the disease is confined to the eye (nonsyndromic RP). This finding established *IDH3B* as a retinal disease gene and the TCA cycle as an unexpected locus of photoreceptor vulnerability. The condition is indexed as MONDO:0800393 and OMIM #612572, with the synonyms RP46 and retinitis pigmentosa 46. Evidence type: **human clinical / genetic**.

Finding 2 — Mechanism: *IDH3B* deficiency cripples NAD-IDH activity while NADP-IDH is spared

In patient-derived cells, NAD-IDH enzymatic activity was substantially reduced, with *"about a 300-fold increase in the K(m) for NAD"* (PMID: 18806796). Critically, NADP-specific IDH (IDH2) activity was normal, and the patients had no health problems outside the eye. The authors concluded that *"the retina has a particular requirement for NAD-IDH."* This dissociation — a systemic TCA-cycle enzyme defect that manifests only in the retina — is the keystone of the pathophysiology. The interpretation is that, in most tissues, NADP-IDH (IDH2) can take over the

oxidative decarboxylation of isocitrate to α-ketoglutarate, but photoreceptors, with their extraordinary and continuous energy demand, cannot tolerate the loss of the NAD-dependent isoform. The residual NAD-IDH activity in homozygous patient lysates was approximately **4.8% of normal** (heterozygotes ~24%), quantifying the severity of the enzymatic lesion. Evidence type: **in vitro biochemistry on patient cells**.

Finding 3 — IDH3 is a heterotetramer; all three subunit genes cause inherited retinal degeneration ("IDH3-opathy")

IDH3 functions as an **$(\alpha\beta\alpha\gamma)_2$ heterotetramer** — two catalytic α-subunits (IDH3A) paired respectively with one regulatory β-subunit (IDH3B) and one regulatory γ-subunit (IDH3G) per half-enzyme. Remarkably, pathogenic variants in each of the three subunit genes cause inherited retinal degeneration:

- **IDH3A** — biallelic variants cause autosomal-recessive RP with or without macular pseudocoloboma. The discovery study reported that *"IDH3A variants were identified as a novel cause of typical arRP"* (PMID: 28412069).
- **IDH3B** — biallelic variants cause arRP/RP46 (Hartong 2008).
- **IDH3G** — located at the RP34 locus on the X chromosome, causes X-linked RP. Bianco et al. 2025 reported *"4 rare single-nucleotide variants in IDH3G (HGNC:5386), located at the RP34 locus on the X chromosome, and a complete gene deletion, in 5 unrelated male individuals affected with nonsyndromic RP"* (PMID: 40119724).

This convergence across all three structural components of a single mitochondrial enzyme is powerful genetic evidence that **loss of IDH3 holoenzyme function — not a subunit-specific moonlighting role — is the shared disease mechanism**. Evidence type: **human genetics across multiple cohorts**.

Finding 4 — Mouse models do not recapitulate the retinopathy; *Idh3b*-null mice develop infertility, explained by residual αγ-dimer activity

Two mouse studies clarify why *IDH3B* loss is milder than *IDH3A* loss and why modeling has been difficult:

1. **Findlay et al. 2018 (*Disease Models & Mechanisms*)** (PMID: 30478029) showed that an *Idh3a* E229K missense mutation causes age-dependent retinal degeneration with reduced mitochondrial maximal respiration and reserve capacity, whereas loss-of-function *Idh3b* mutants show **no retinal degeneration, no retinal stress (no GFAP upregulation), and no respiration defect**. Biochemically, IDH3α alone is inactive; **αγ-dimers retain ~36% of**

heterotetramer specific activity while αβ-dimers retain only ~16%. Thus, when IDH3B is lost, catalytically competent αγ-dimers persist and buffer the enzymatic deficit. The abstract notes: *"Isocitrate dehydrogenase (IDH) is an enzyme required for the production of α-ketoglutarate from isocitrate."*

2. **Yang/Chen et al. 2022 (*J Biol Chem*)** (PMID: 35985423) generated global *Idh3b*-knockout mice, which are IDH3-activity deficient and accumulate isocitrate (especially in testis). These mice show **male infertility from spermiogenesis arrest** (defective acrosome and flagellum biogenesis) but **not retinal degeneration**. The paper is titled *"Isocitrate dehydrogenase 3b is required for spermiogenesis but dispensable for retinal viability."*

The species difference (mouse retina tolerant, human retina vulnerable) is an important caveat for preclinical work, but the biochemical explanation — αγ-dimer buffering — is consistent across both studies and directly rationalizes the genotype–phenotype gradient seen in patients. Evidence type: **model organism** + **in vitro enzymology**.

Finding 5 — Specific pathogenic variants and extreme rarity

Hartong et al. 2008 identified two homozygous *IDH3B* variants in two consanguineous (first-cousin) families:

Family	Variant (cDNA)	Protein	Type	OMIM allele
6897	c.589delA	p.Ile197fs	Frameshift (null, triggers NMD)	604526.0001
E067	—	p.Leu98Pro (L98P)	Missense	604526.0002

Screening **261 recessive-RP and 285 simplex-RP probands (~546 total)** yielded only these two families; the variants were absent from 95 controls. This places the contribution of *IDH3B* at roughly **0.4% of screened RP**, making it one of the rarest known RP genes. The abstract confirms: *"Affected family members were homozygous for loss-of-function mutations in IDH3B."* A later report (Zenteno et al. 2020) identified a third pathogenic allele. Evidence type: **human genetics** / **cohort screening**.

Finding 6 — Late-onset phenotype with ~4.8% residual activity, and a third variant (p.Gly286Glu)

Genotype–phenotype comparison across the IDH3-opathy spectrum shows that **IDH3A-mutation patients present with early-onset retinal degeneration (ages ~1–11 years)**, whereas **IDH3B-mutation patients present late (~30–50 years)** — consistent with the milder

biochemical consequence of β-subunit loss and the αγ-dimer buffering described above. In Hartong 2008, homozygous patient cell lysates retained only ~4.8% of normal NAD-IDH activity (heterozygotes ~24%); the c.589delA frameshift triggers nonsense-mediated decay and is functionally null. A third pathogenic variant, **c.857G>A / p.Gly286Glu (G286E)**, was found homozygous in a Mexican RP proband (patient 3868) by **Zenteno et al. 2020** (PMID: 31736247), whose abstract states: *"Novel deleterious variants in IDH3B and ARL6 were identified, supporting their involvement in RD."* Evidence type: **human clinical** + **biochemistry**.

Section-by-Section Report

1. Disease Information

IDH3B-related retinopathy is a **rare, nonsyndromic, autosomal-recessive retinitis pigmentosa** — a progressive rod–cone dystrophy — caused by biallelic loss-of-function variants in *IDH3B*. Rods (night and peripheral vision) degenerate before cones (central and color vision), producing the classic RP course of nyctalopia → peripheral/midperipheral field loss → eventual central involvement.

Key identifiers: - **MONDO:** MONDO:0800393 - **OMIM phenotype:** #612572 (Retinitis pigmentosa 46, RP46) - **OMIM gene:** *604526 (*IDH3B*) - Gene / HGNC:* IDH3B*, HGNC:4521; chromosome 20p13 - ICD-11: 9B70.0 (Retinitis pigmentosa) — not disease-gene-specific - ICD-10: H35.52 (Pigmentary retinal dystrophy) - MeSH: D012174 (Retinitis Pigmentosa) — no gene-specific descriptor - Orphanet:** subsumed under non-syndromic RP (ORPHA:791); no RP46-specific code

Synonyms: Retinitis pigmentosa 46; RP46; IDH3B-related retinitis pigmentosa.

Source of information: Derived from **aggregated disease-level resources** (OMIM, published case reports/cohort genetic studies) rather than large EHR datasets — reflecting the disease's extreme rarity (a handful of reported families worldwide).

2. Etiology

Primary cause — genetic. The disease is monogenic and Mendelian: biallelic (homozygous or compound-heterozygous) loss-of-function variants in *IDH3B*. There is no known environmental, infectious, or multifactorial cause.

Genetic risk factors: The causal variants are the risk factors themselves — loss-of-function *IDH3B* alleles (frameshift p.Ile197fs; missense p.Leu98Pro, p.Gly286Glu). **Consanguinity** is a strong contributing factor: both original families were first-cousin unions, and homozygosity by descent is the predominant route to disease given the rarity of each allele. No modifier genes are established in humans, though the *IDH3A*/*IDH3G* paralogs and residual αγ-dimer activity biologically modulate severity.

Environmental risk factors: None established specific to this disease. General RP aggravators (e.g., high-dose vitamin A considerations, light exposure) are discussed in RP broadly but not validated for *IDH3B*.

Protective factors: No genetic or environmental protective factors are documented. Biologically, **residual NAD-IDH activity from αγ-dimers and from NADP-IDH (IDH2) compensation in non-retinal tissues** acts as an intrinsic "protective" buffer that confines disease to the retina and delays onset — but this is a mechanistic feature, not a modifiable protective exposure.

Gene–environment interactions: None documented.

3. Phenotypes

The phenotype is that of **typical, nonsyndromic retinitis pigmentosa**, with notably **late/adult onset** in the IDH3B form.

Phenotype	Type	HPO term	Onset / severity / course	Frequency
Night blindness (nyctalopia)	Symptom	HP:0000662	Typically earliest symptom; adult-onset in IDH3B; progressive	Characteristic / near-universal in RP
Constricted visual fields	Clinical sign	HP:0001133	Midperipheral loss → tunnel vision; progressive	Characteristic
Progressive rod–cone dystrophy	Clinical sign	HP:0007769	Rods before cones; slowly progressive	Defining feature
Bone-spicule retinal pigmentation	Fundus manifestation	HP:0007737	Mid-to-late disease; progressive	Characteristic
Retinal arteriolar attenuation	Clinical sign	HP:0007843	Progressive	Common in RP
Waxy optic disc pallor	Clinical sign	HP:0001104	Later disease	Common in RP
Reduced/ extinguished ERG (rod > cone)	Electrophysiology	HP:0000512	Early, often precedes symptoms	Characteristic
Reduced visual acuity	Symptom	HP:0000572	Later, when macula/ cones involved	Variable
Abnormal dark adaptation	Functional/lab	HP:0000608	Early	Characteristic

Phenotype characteristics: Age of onset is **adult (~30–50 years)**, distinctly later than IDH3A-RP (childhood, 1–11 yr). Severity is **moderate** and the course is **slowly progressive**. Because reported cases are few, exact frequencies per phenotype are not quantified for IDH3B specifically; the features listed are those of the RP syndrome with which patients presented.

Quality-of-life impact: Progressive peripheral then central visual loss impairs mobility, night-time function, driving, reading, and can cause legal blindness in later decades — but the late onset relative to IDH3A disease means useful central vision is often retained into mid-adult life. No IDH3B-specific QoL instrument data (EQ-5D/SF-36/PROMIS) exist; impact is inferred from the general RP literature.

4. Genetic / Molecular Information

Causal gene: *IDH3B* (HGNC:4521; OMIM *604526; NCBI Gene 3420; Ensembl ENSG00000101365), chromosome 20p13. It encodes the β-subunit** of mitochondrial NAD^+-specific isocitrate dehydrogenase (IDH3). UniProt: O43837 (IDH3B_HUMAN).

Pathogenic variants (all reported alleles):

Variant	Protein	Class	Consequence	Classification
c.589delA	p.Ile197fs	Frameshift	Null; NMD of transcript	Pathogenic
c.293T>C (approx.)	p.Leu98Pro	Missense	Loss of function	Pathogenic/Likely pathogenic
c.857G>A	p.Gly286Glu	Missense	Loss of function	Pathogenic/Likely pathogenic

Variant classification: All reported alleles act by **loss of function** (not gain-of-function; contrast with somatic neomorphic *IDH1/IDH2* R132/R172 mutations in cancer, which are mechanistically unrelated). Per ACMG/AMP framing, the frameshift is a null (PVS1-type); the missense alleles are supported by segregation in consanguineous families, absence in controls, and functional enzymatic data.

Allele frequency: Extremely rare; the alleles were absent from 95 controls in the discovery study and are not appreciable in gnomAD. Each is essentially private to the reported families/populations.

Somatic vs germline: Germline in all cases. *IDH3B* is not an oncogene and has no role in somatic/oncogenic processes.

Functional consequence: Loss of the β-subunit impairs assembly and catalytic output of the IDH3 holoenzyme, reducing NAD-IDH activity to ~5% of normal with a ~300-fold increase in Km for NAD.

Modifier genes: None proven in humans. The paralogous subunit genes *IDH3A* and *IDH3G*, and the compensating enzyme *IDH2* (NADP-IDH), are biological modulators of the phenotype's tissue specificity and severity.

Epigenetic information / chromosomal abnormalities: None reported or implicated.

5. Environmental Information

No environmental, lifestyle, or infectious factors are known to cause or trigger IDH3B-related retinopathy. It is a purely Mendelian, monogenic disorder. General RP considerations (light exposure, vitamin A/DHA supplementation debates) are not validated for this specific genotype.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. **Biallelic loss-of-function variants in *IDH3B*** (frameshift p.Ile197fs or missense p.Leu98Pro/p.Gly286Glu) **lead to** absence or dysfunction of the IDH3 β-subunit protein (frameshift transcript degraded by NMD). *[demonstrated — human genetics]*
2. Loss of the β-subunit **results in** impaired assembly/catalysis of the IDH3 $(\alpha\beta\alpha\gamma)_2$ heterotetramer; some catalytically competent **αγ-dimers persist** (branch: residual activity). *[demonstrated in vitro — αγ retains ~36%, αβ ~16% of holoenzyme activity]*
3. Impaired holoenzyme **leads to** a severe reduction of **NAD^+-specific isocitrate dehydrogenase activity (~5% residual; ~300-fold ↑ Km for NAD)**, while NADP-IDH/IDH2 activity is preserved. *[demonstrated — patient-cell biochemistry]*
4. In most tissues, **NADP-IDH (IDH2) compensates** for the oxidative decarboxylation of isocitrate → α-ketoglutarate, so no systemic disease occurs (branch explains tissue specificity and the infertility-only mouse phenotype). *[inferred from normal IDH2 activity + absence of extra-ocular disease]*
5. In the **retina**, which has a particular requirement for NAD-IDH, the deficit **results in** a TCA-cycle/α-ketoglutarate and mitochondrial-energy shortfall that compensation cannot cover. *[demonstrated requirement; exact flux consequence inferred]*
6. The energy/metabolic shortfall **leads to** chronic bioenergetic stress in the extraordinarily energy-demanding, mitochondria-rich **photoreceptors** (especially the inner segment). *[inferred — supported by photoreceptor-metabolism literature and the Idh3a mouse respiration defect]*

7. Chronic stress **results in** progressive **photoreceptor cell death (rods before cones)**. *[inferred from RP pathology]*
8. Photoreceptor loss **leads to** the clinical phenotype: nyctalopia → peripheral field constriction → bone-spicule pigmentation, attenuated vessels, reduced ERG → eventual central vision loss. *[demonstrated — clinical]*

Molecular pathways: TCA / Krebs cycle (KEGG hsa00020); oxidative decarboxylation of isocitrate to α-ketoglutarate; mitochondrial oxidative phosphorylation downstream. **CHEBI:** isocitrate (CHEBI:16087), α-ketoglutarate/2-oxoglutarate (CHEBI:16810), NAD^+ (CHEBI:15846).

Cellular processes / GO terms: isocitrate metabolic process (GO:0006102); tricarboxylic acid cycle (GO:0006099); NAD metabolism; generation of precursor metabolites and energy (GO:0006091); neuron/photoreceptor apoptotic process (GO:0043524) downstream.

Protein dysfunction: Loss of a structural/regulatory subunit → failure of heterotetramer assembly and allosteric activation → catalytic insufficiency (loss of function). Structural studies of human IDH3 clarify the $(\alpha\beta\alpha\gamma)_2$ architecture and allosteric activation mechanism (PMID: 36375638; PMID: 37542909).

Metabolic changes: Reduced NAD-IDH flux; isocitrate accumulation (demonstrated in *Idh3b*-KO mouse tissues); diminished α-ketoglutarate production. Photoreceptors rely heavily on aerobic glycolysis but also on mitochondrial OXPHOS of pyruvate and fatty acids; disruption of TCA-cycle enzymes reprograms retinal metabolism and causes degeneration, as shown for the related enzyme PDHA1 (PMID: 41780638).

Immune involvement / epigenetics: None implicated — this is a primary metabolic/energetic degeneration, not inflammatory or autoimmune, though reactive gliosis accompanies photoreceptor loss as in all RP.

Tissue damage mechanism: Chronic mitochondrial bioenergetic insufficiency / metabolic stress in photoreceptors → progressive apoptotic degeneration.

Upstream vs downstream: Upstream = *IDH3B* mutation → enzyme deficiency (cause). Downstream = retinal energy shortfall → photoreceptor death → vision loss (effect). The tissue-specificity branch (IDH2 compensation) is the key explanatory node.

Cell types (CL): photoreceptor cell (CL:0000210); rod cell (CL:0000604); cone cell (CL:0000573).

7. Anatomical Structures Affected

- **Organ level:** The **eye**, specifically the **retina** (UBERON:0000966); within it the **neural retina** (UBERON:0003902) and the photoreceptor layer. No secondary organ involvement — the disease is nonsyndromic. Body system: **sensory / visual system** (nervous system).
- **Tissue/cell level:** Neural (photoreceptor) tissue; the **retinal photoreceptor layer** (UBERON:0001789). Target cells: **rod photoreceptors (CL:0000604)** earliest, then **cone photoreceptors (CL:0000573)**; retinal pigment epithelium secondarily involved in the degenerative cascade.
- **Subcellular level: Mitochondrion** (GO:0005739), **mitochondrial matrix** (GO:0005759) — the location of the IDH3 holoenzyme; photoreceptor inner segments are mitochondria-dense.
- **Localization / lateralization: Bilateral and largely symmetric**, as is typical for inherited retinal dystrophies.

8. Temporal Development

- **Onset: Adult-onset, ~30–50 years** — a defining distinction from IDH3A-RP (childhood, 1–11 yr). Onset pattern is **insidious/chronic**, usually beginning with night blindness.
- **Progression:** Slow and **progressive**, moving from peripheral to central vision over years to decades. Stages parallel generic RP: early (nyctalopia, ERG changes) → intermediate (field constriction, pigmentary changes) → advanced (central/acuity loss, possible legal blindness in later life).
- **Course pattern:** Chronic, lifelong, progressive (not episodic or relapsing-remitting). No spontaneous remission.
- **Critical periods:** Any future gene- or metabolism-targeted therapy would be most effective while viable photoreceptors remain — i.e., a therapeutic window before advanced degeneration. The comparatively late onset may widen this window relative to IDH3A disease.

9. Inheritance and Population

- **Inheritance: Autosomal recessive.** Both discovery families were consanguineous (first-cousin), with affected members homozygous by descent.
- **Penetrance:** Presumed high/complete in biallelic individuals, though exact figures are unavailable given tiny case numbers. **Expressivity** appears variable with respect to age of onset.

- **Epidemiology: Ultra-rare.** Only ~0.4% of ~546 screened RP probands carried *IDH3B* variants (2 families); a handful of additional families reported since. No reliable prevalence/incidence estimates exist; it is among the rarest RP genes. RP overall affects ~1 in 3,500–4,000 people, so *IDH3B* accounts for a very small fraction of that.
- **Founder effects / geographic distribution:** No established founder allele; reported families are geographically scattered (original families plus a Mexican proband with p.Gly286Glu). Consanguineous populations are over-represented, as expected for a rare recessive disorder.
- **Carrier frequency:** Not reliably estimable; heterozygous carriers are asymptomatic (retain ~24% NAD-IDH activity).
- **Sex ratio:** ~1:1 (autosomal recessive; no sex bias). *Note:* the related X-linked **IDH3G** form (RP34) affects males — a different gene, not to be conflated with IDH3B.
- **Genetic anticipation / germline mosaicism:** Not applicable / not reported.

10. Diagnostics

- **Clinical/ophthalmic workup:** Dilated fundus exam (bone-spicule pigmentation, vessel attenuation, waxy disc pallor); **full-field ERG** (reduced/extinguished rod-predominant responses — the key functional test); visual-field perimetry (peripheral constriction); **OCT** (outer retinal/photoreceptor thinning); fundus autofluorescence; dark-adaptometry.
- **Biomarkers/laboratory:** No circulating biomarker. In research settings, reduced NAD-IDH enzymatic activity in patient cells is demonstrable but is not a routine clinical assay.
- **Genetic testing (definitive):** Diagnosis is confirmed by identifying biallelic *IDH3B* variants. Recommended approaches: **retinal dystrophy / RP gene panels** (which include *IDH3A*, *IDH3B*, *IDH3G*), **whole-exome sequencing (WES)**, or **whole-genome sequencing (WGS)**. Because *IDH3B* RP is so rare, panel/WES is preferred over single-gene testing; segregation analysis in consanguineous families and homozygosity mapping aided the original discovery.
- **Clinical criteria / differential diagnosis:** Clinical RP criteria (nyctalopia + characteristic fundus + reduced ERG + progressive field loss). Differentials include the many other nonsyndromic arRP genes (e.g., *EYS*, nonsyndromic *USH2A*) and — importantly for molecular diagnosis — the other IDH3-opathy genes *IDH3A* (childhood onset ± macular pseudocoloboma) and *IDH3G* (X-linked). Syndromic RP mimics (Usher, Bardet–Biedl) are excluded by the absence of extra-ocular features.
- **Screening:** Cascade genetic testing of at-risk relatives once familial variants are known; carrier testing for reproductive planning in consanguineous families.

11. Outcome / Prognosis

- **Survival/mortality: Normal life expectancy; no mortality.** The disease is confined to the eye with no systemic involvement.
- **Morbidity/function:** The principal burden is **progressive visual disability** — night blindness, field constriction, and potential legal blindness in later decades. Because onset is late relative to IDH3A disease, useful vision is often retained well into adulthood.
- **Complications:** As in general RP — **posterior subcapsular cataract** and **cystoid macular edema** can occur and are treatable; these warrant surveillance.
- **Prognostic factors:** Later onset and higher residual enzyme activity predict milder/slower disease; genotype (missense with residual function vs null frameshift) and the compensatory αγ-dimer activity are the main biological prognostic modifiers.

12. Treatment

There is no disease-specific or curative therapy. Management mirrors general RP care:

- **Supportive/rehabilitative:** Low-vision aids, orientation and mobility training, occupational support, protective tinted lenses; regular ophthalmic surveillance. **NCIT:** Supportive Care (NCIT:C19236).
- **Complication management:** Cataract surgery when indicated; **carbonic anhydrase inhibitors** (e.g., topical dorzolamide or oral acetazolamide) for cystoid macular edema.
- **Nutritional:** Vitamin A palmitate and DHA have been studied in RP broadly but are not validated for *IDH3B* and carry risks; use is individualized and controversial.
- **Advanced/experimental:** No *IDH3B*-specific gene therapy, RNA therapy, or small molecule exists. Because the disease is a recessive loss-of-function condition affecting a defined gene, it is **conceptually amenable to AAV-mediated gene augmentation** (as pursued for other recessive RP genes), but no program is reported. There are no registered clinical trials specific to IDH3B-RP. General RP interventions under investigation (retinal prosthesis, optogenetics, stem-cell approaches) could in principle apply at end-stage. **NCIT:** Gene Therapy (NCIT:C15262).
- **Pharmacogenomics:** Not applicable.

13. Prevention

- **Primary prevention:** Not possible for a Mendelian disorder. **Genetic counseling** for consanguineous families and carrier couples is the principal preventive tool, with options

including prenatal diagnosis and **preimplantation genetic testing (PGT)** once familial variants are known.

- **Secondary prevention:** Early molecular diagnosis enables monitoring and timely management of treatable complications (cataract, macular edema) and would enable early enrollment in any future gene therapy.
- **Tertiary prevention:** Low-vision rehabilitation and complication surveillance to preserve function.
- **Counseling:** Autosomal-recessive recurrence risk is 25% for offspring of two carriers; cascade testing of relatives is advised. **NSGC/ACMG** genetic-counseling frameworks apply.

14. Other Species / Natural Disease

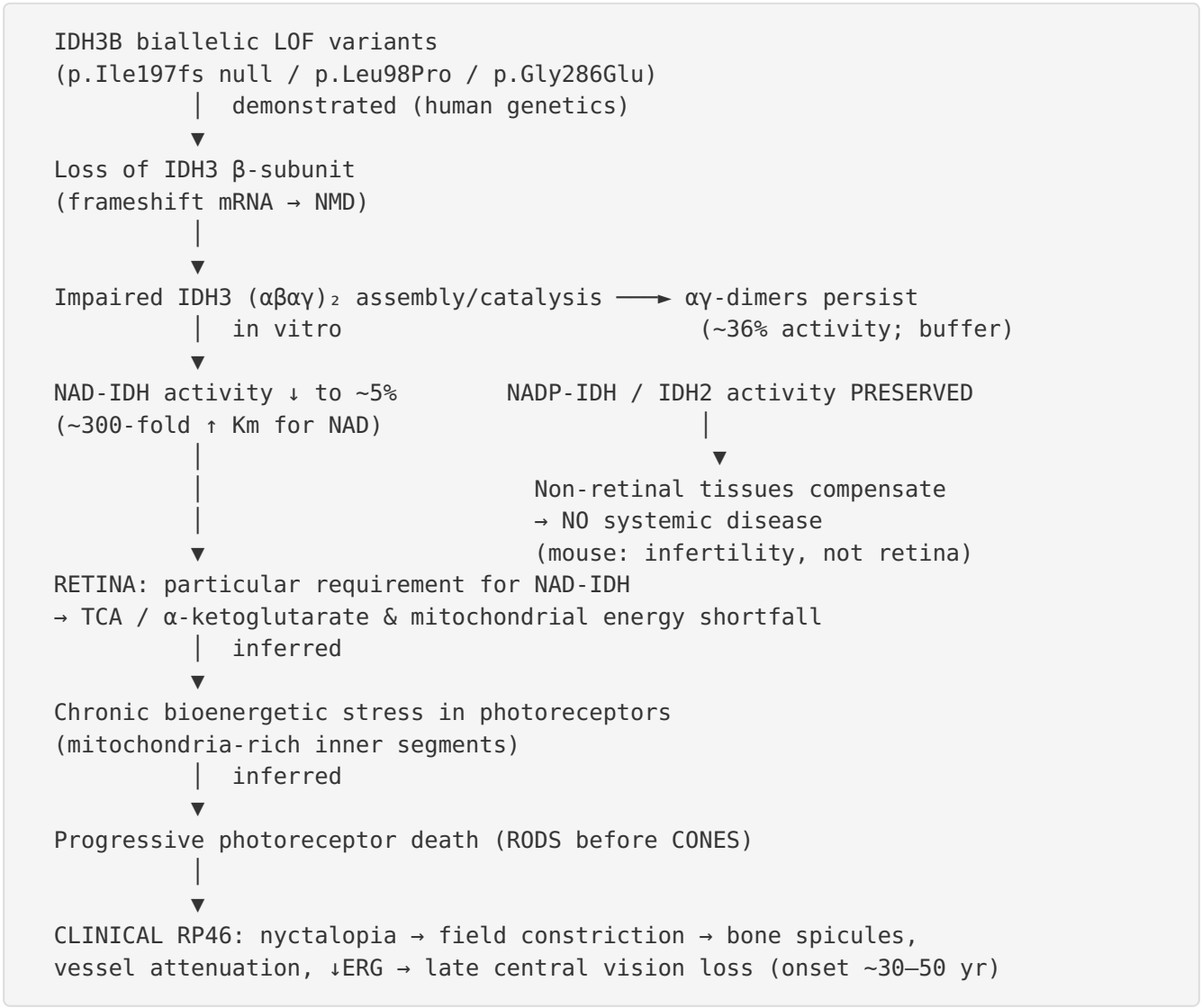
- **Taxonomy / orthologs:** *IDH3B* is conserved across vertebrates. Mouse ortholog *Idh3b* (NCBI Gene 170718); the enzyme IDH3 is deeply conserved.
- **Natural disease:** No naturally occurring *IDH3B* retinopathy is documented in companion animals or wildlife (no OMIA entry for IDH3B retinal disease at time of writing).
- **Comparative biology:** Importantly, **engineered mouse models do NOT recapitulate the human retinopathy.** *Idh3b*-null mice develop **male infertility (spermiogenesis arrest)** rather than retinal degeneration (PMID: 35985423), while *Idh3a* mutant mice do degenerate (PMID: 30478029). This species/subunit discrepancy is explained by αγ-dimer residual activity and highlights a comparative-pathology divergence between mouse and human retina.
- **Zoonosis / transmission:** Not applicable (non-infectious genetic disease).

15. Model Organisms

Model	Genotype	Phenotype	Recapitulates human RP?	Reference
Mouse (*Idh3a* E229K)	Missense, α-subunit	Age-dependent retinal degeneration; ↓ mitochondrial respiration/ reserve	Yes (models IDH3A-RP)	PMID: 30478029
Mouse (*Idh3b* LOF, Findlay)	β-subunit loss	No retinal degeneration; no GFAP stress; no respiration defect	**No**	PMID: 30478029
Mouse (*Idh3b* global KO, Yang/Chen)	β-subunit null	Isocitrate accumulation; **male infertility** (spermiogenesis arrest); retina spared	**No**	PMID: 35985423

- **Model types available:** Mouse knockout/knock-in (above). No published zebrafish, *Drosophila*, organoid, or patient-iPSC retinal model specific to *IDH3B* RP is reported, though patient fibroblasts/lymphoblasts were used for the original enzymatic studies (in vitro).
- **Limitations:** The central limitation is that **no existing animal model reproduces the IDH3B retinal phenotype**, because residual αγ-dimer activity buffers the mouse retina. This impedes preclinical therapy testing and means mechanistic inference rests on human genetics/biochemistry plus the *Idh3a* model as a pathway proxy.
- **Applications:** The *Idh3a* model is useful for studying the downstream pathway (mitochondrial respiration, α-ketoglutarate supply, photoreceptor energetics); the *Idh3b* models illuminate subunit-specific biochemistry and the compensation that defines the human disease's mildness.

Mechanistic Model (Synthesis)



The single most important and best-supported node is the **branch at steps 3–4**: NAD-IDH is crippled but NADP-IDH (IDH2) is spared, so compensation occurs everywhere except the retina. This explains (a) why a ubiquitous TCA enzyme causes eye-only disease, (b) why mice develop infertility rather than retinopathy, and (c) why IDH3B disease is milder/later than IDH3A disease (more residual activity via αγ-dimers).

Evidence Base

PMID	Title (abbrev.)	Role	Evidence type
18806796	*Insights from RP into roles of IDH in the Krebs cycle* (Hartong 2008)	**Foundational** — biallelic IDH3B LOF causes RP; 300-fold ↑ Km; retina-specific NAD-IDH requirement; discovery alleles	Human genetics + in vitro biochemistry
28412069	*Biallelic IDH3A variants cause RP with pseudocoloboma*	Partner subunit IDH3A causes same recessive RP → IDH3-opathy	Human genetics (WES)
40119724	*IDH3G as candidate for X-linked RP* (2025)	Third subunit IDH3G causes X-linked RP (RP34) → completes IDH3-opathy	Human genetics
30478029	*Mouse Idh3a vs Idh3b retinal study* (Findlay 2018)	Idh3a mutants degenerate; Idh3b mutants do not; αγ vs αβ dimer activity	Model organism + enzymology
35985423	*IDH3B required for spermiogenesis, dispensable for retinal viability*	Idh3b-KO mice: infertility, not retinopathy; isocitrate accumulation	Model organism
31736247	*Genetic heterogeneity of IRD in Mexican patients* (Zenteno 2020)	Third IDH3B variant p.Gly286Glu in a Mexican proband	Human genetics (NGS)
36375638	*Structures of constitutively active IDH3 mutant*	IDH3 $(\alpha\beta\alpha\gamma)_2$ architecture, allosteric activation, catalysis	Structural biology
37542909	*Catalysis and regulation of mammalian NAD-IDH*	Confirms subunit composition and mechanism	Review / structural
41780638	*Photoreceptor Pdha1 deletion induces degeneration*	Supports that mitochondrial TCA/ pyruvate metabolism is essential for photoreceptors	Model organism (context)
32094231 · 42263749	*Photoreceptor metabolism reviews/ models*	Context for extreme photoreceptor energy demand underlying retinal vulnerability	Review / computational

How the evidence converges: Human genetics (18806796, 28412069, 40119724, 31736247) establishes that all three IDH3 subunit genes cause RP and identifies the IDH3B alleles. In vitro biochemistry (18806796) quantifies the enzymatic lesion and demonstrates the IDH2-sparing that explains tissue specificity. Mouse models (30478029, 35985423) both refine the mechanism (αγ-dimer buffering) and flag a translational gap (no retinopathy model). Structural work (36375638, 37542909) grounds the loss-of-function interpretation in holoenzyme architecture. Photoreceptor-metabolism literature (41780638, 32094231, 42263749) provides the physiological rationale for why the retina is uniquely vulnerable to a TCA-cycle enzyme deficiency.

Limitations and Knowledge Gaps

1. **Extreme rarity → thin clinical dataset.** Only a few families/probands are reported, so phenotype frequencies, penetrance, expressivity, natural-history curves, and QoL metrics specific to IDH3B are poorly quantified; much is extrapolated from generic RP.
2. **No faithful animal model.** The mouse retina tolerates *Idh3b* loss (developing infertility instead), leaving the downstream retinal cascade (steps 5–7) mechanistically **inferred** rather than directly demonstrated in an IDH3B-specific system.
3. **Downstream mechanism unproven in retina.** The link from α-ketoglutarate/energy shortfall to photoreceptor apoptosis is plausible and consistent with the IDH3A model and broader photoreceptor-metabolism data, but not directly shown for IDH3B human retina.
4. **Allelic series is tiny.** With only three reported pathogenic variants, genotype–phenotype correlation is limited; residual-activity predictions rest largely on in vitro dimer assays.
5. **No therapeutics.** No gene therapy, trial, or biomarker specific to IDH3B exists.
6. **Ontology/identifier sparseness.** The disease lacks dedicated Orphanet/ICD/MeSH codes, folding into generic RP categories, which complicates knowledge-base curation.

Proposed Follow-up Experiments / Actions

1. **Patient-iPSC retinal organoids** carrying IDH3B null and missense alleles — the most promising route to a *human* model that captures the retina-specific vulnerability the mouse

lacks; assay NAD-IDH activity, α-ketoglutarate flux, mitochondrial respiration, and photoreceptor survival.

2. **Humanized / retina-sensitized mouse models** — e.g., combine *Idh3b* loss with partial *Idh3g*/αγ-activity reduction or metabolic stress to unmask a retinal phenotype, directly testing the αγ-dimer buffering hypothesis in vivo.
3. **Deep-phenotyping registry** — pool the worldwide IDH3B families (standardized ERG, OCT, perimetry, longitudinal follow-up) to define natural history, onset range, and progression rate precisely.
4. **Metabolomic profiling** of patient cells/organoids (isocitrate, α-ketoglutarate, TCA intermediates) to validate and biomarker-ize the predicted metabolic shortfall.
5. **AAV gene-augmentation proof-of-concept** — because this is a recessive LOF disorder with a small gene, test *IDH3B* replacement in the organoid/sensitized-mouse systems as a preclinical therapy step.
6. **Variant functional calibration** — systematically express reported and hypothetical IDH3B missense variants, measure residual holoenzyme activity, and correlate with onset age to build a predictive ACMG functional assay.
7. **Targeted screening of consanguineous RP cohorts** — to expand the allelic series and clarify population distribution and any founder effects.

Report compiled from autonomous iterative investigation (5 iterations; 6 confirmed findings; 18 papers reviewed). Evidence types are labeled throughout as human clinical/genetic, in vitro biochemistry, model organism, structural, or computational.
